ECONOMIC GROWTH
SPRING 2020
EXERCISE 2

Shorter questions

1. Piketty and Saez documents that income inequality has exhibited a distinct path over 20th century. What is the pattern, and how do the authors propose that the path may be explained?

2. Krusell and Smith provide a critique of one of Thomas Piketty's theories of inequality. Which theory? Provide a short recap of the critique. Does their critique mean Piketty is nessesarily wrong?

3. Krusell and Smith (2015), in discussing Thomas Piketty's theories, argues (p. 736): "*Optimal-savings theory implies, more generally, that on a balanced growth path, both the gross and net saving rates depend positively on g.*" Here "g" refers to the productivity growth rate. Do you agree or disagree? Explain.

Analytical questions

Consider an economy in the process of development. Consumer i faces the following problem

$$\max_{\{c_{it}\}_0^\infty} \int_0^\infty \ln(c_{it})\, e^{-\rho t} dt$$

where c_i is consumption of consumer i, and ρ is the rate of discount. As usual, consumption is required to be non-negative. The flow constraint for wealth of consumer i k_{it} is

$$\dot{k}_{it} = (r - \tau)\, k_{it} + w_t + \theta_t - c_{it},\ k_{i0} \text{ given}$$

where r is the (as it turn out, constant) real rate of interest, τ is a time constant wealth tax, w_t is the real wage, and θ_t is a lump sum transfer funded by the government. Finally, the consumer is constrained by the requirement, than he or she has to be solvent in the long run, $t \to \infty$. Agents in the economy have different amounts of wealth, but they all have one unit of labor which is supplied inelastically. The total labor force, L, is constant, and the aggregate capital stock is $\sum_i k_{it} = K_t$. Capital depreciation is ignored ($\delta = 0$). The government raises wealth taxes, and uses the proceeds to fund the lump sum transfer; the budget is balanced. Accordingly,

$$\tau K_t = \theta_t L.$$

1. Solve the consumer problem, and show that a nessesary condition for optimality is that the consumer abides by the consumption euler

$$\frac{\dot{c}_{it}}{c_{it}} = r - \tau - \rho.$$

2. The representative firm operates the production function $Y_t = K_t^{\alpha} (A_t L)^{1-\alpha}$. Factor markets are competitive, and the firm takes A_t as given. Show that the first order conditions for profitmaximization implies

$$r = \alpha \frac{Y_t}{K_t}, w_t = (1-\alpha) \frac{Y_t}{L}$$

3. It is assumed that productivity evolves due to (external) learning-by-investing. Hence, $A_t = \lambda K_t$, $\lambda > 0$. Does the model allow for endogenous growth?

4. Show that the equilibrium real rate of interest, and wage rate, is given by

$$r = \alpha \frac{Y}{K} = \alpha (\lambda L)^{1-\alpha}, \ w = (1-\alpha)(\lambda L)^{1-\alpha} K_t / L$$

and explain why the growth rate of output can be written

$$\gamma = \alpha (\lambda L)^{1-\alpha} - \tau - \rho.$$

5. Show that the consumption path of each individual in the economy is given by

$$c_{it} = \left[\left((1-\alpha)(\lambda L)^{1-\alpha} + \tau \right) \sigma_i + \rho \right] k_{it},$$

where $\sigma_i \equiv \frac{K_t/L}{k_{it}}$.

6. Show that the problem of a consumer, of chosing his or her preferred *time constant* wealth tax τ at time zero, can be written (*hint:* since we are on a balanced growth path, $c_{it} = c_{i0} e^{\gamma t}$. Use this information to solve the integral $\int_0^{\infty} \ln c_{it} e^{-\rho t} dt$):

$$\max_{\tau} \frac{1}{\rho} \left(\ln c_{i0} + \frac{\gamma}{\rho} \right)$$

s.t.

$$c_{i0} = \left[\left((1-\alpha)(\lambda L)^{1-\alpha} + \tau \right) \sigma_i + \rho \right] k_{i0}$$

$$\gamma = \alpha (\lambda L)^{1-\alpha} - \tau - \rho$$

7. Solve the consumer's problem. Suppose the tax rate is chosen through majority voting. Which tax rate will be selected?

8. In the present economy, the growth rate selected by a social planner would be

$$\gamma = (\lambda L)^{1-\alpha} - \rho.$$

Can this solution be the outcome from majority voting? Why/Why not?